

- 3 (a) (i) stress = force / (cross-sectional) area B1 [1]
- (ii) strain = extension / original length **or** change in length / original length B1 [1]
- (b) point beyond which material does not return to the original length / shape / size when the load / force is removed B1 [1]

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- (c) UTS is the maximum force / original cross-sectional area M1
 wire is able to support / before it breaks A1 [2]

allow one: maximum stress the wire is able to support / before it breaks

- (d) (i) straight line from (0,0) M1
 correct shape in plastic region A1 [2]

- (ii) only a straight line from (0,0) B1 [1]

- (e) (i) ductile: initially force proportional to extension then a large extension for small change in force B1
 brittle: force proportional to extension until it breaks B1 [2]

- (ii) 1. does not return to its original length / permanent extension (as entered plastic region) B1
 2. returns to original length / no extension (**as no plastic region / still in elastic region**) B1 [2]

- 4 (a) electric field strength = force / positive charge B1 [1]